

BAYESUVIUS

A VISUAL DICTIONARY OF BAYESIAN
NETWORKS AND CAUSAL INFERENCE



ROBERT R. TUCCI

Bayesuvius,
a visual dictionary of Bayesian Networks and
Causal Inference

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This book is constantly being expanded and improved. To download the latest version, go to <https://github.com/rrtucci/Bayesuvius>

Bayesuvius

by Robert R. Tucci

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Figure 1: View of Mount Vesuvius from Pompeii



Figure 2: Mount Vesuvius and Bay of Naples

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Prologues

Chapter 93

Renormalization Group for Bayesian Networks

In the previous chapter (Ch. 92), we discussed the Renormalization Group (RG) in general. In this chapter, we discuss how to apply the RG to bnets.

As an example, we apply the RG techniques for bnets to a dynamical Bayesian network (dbnet) (see Ch. 28) of our own devising, that is designed so as to incorporate all the physics of the standard Ising model (the latter is an undirected DAG (Markov model) instead of a DAG (bnet)). We have implemented our Ising-dbnet in the open source software `bnet_rg` (see Ref.[94]).

This chapter and the software `bnet_rg` are original research, first published here.

93.1 Preliminaries

In this section we remind the reader of some theory (on 1. Information theory 2. Standard Ising Model) that will be used to discuss the main topic of this chapter.

93.1.1 Information Theory

Recall that in Shannon Information theory (see Ch. 99), we define for random variables $\underline{x}, \underline{y}$

- the **entropy**

$$H(\underline{x}) = - \sum_{x \in \text{val}(\underline{x})} P(x) \ln P(x) \quad (93.1)$$

- the **joint entropy**

$$H(\underline{x}, \underline{y}) = - \sum_{x \in \text{val}(\underline{x})} \sum_{y \in \text{val}(\underline{y})} P(x, y) \ln P(x, y) \quad (93.2)$$

- the **conditional information**

$$H(\underline{y} \mid \underline{x}) = - \sum_{x \in \text{val}(\underline{x})} \sum_{y \in \text{val}(\underline{y})} P(x, y) \ln P(y \mid x) \quad (93.3)$$

$$= H(\underline{x}, \underline{y}) - H(\underline{x}) \quad (93.4)$$

- the **mutual information (MI)**

$$H(\underline{x} : \underline{y}) = \sum_{x \in \text{val}(\underline{x})} \sum_{y \in \text{val}(\underline{y})} P(x, y) \ln \frac{P(x, y)}{P(x)P(y)} \quad (93.5)$$

$$= H(\underline{y}) - H(\underline{y} \mid \underline{x}) = H(\underline{x}) - H(\underline{x} \mid \underline{y}) \quad (93.6)$$

Here are some standard inequalities that we will find useful in the rest of this chapter.

Claim 139

$$0 \leq H(\underline{x}) \leq \ln |\text{val}(\underline{x})| \quad (93.7)$$

$$0 \leq H(\underline{y} \mid \underline{x}) \leq H(\underline{x}, \underline{y}) \quad (93.8)$$

$$0 \leq H(\underline{x} : \underline{y}) \leq \min(H(\underline{x}), H(\underline{y})) \quad (93.9)$$

proof: Left to the reader.

QED

We will use MI to measure the correlation between random variables $\underline{x}$ and $\underline{y}$.

Note that MI achieves its zero lower bound when $P(x, y) = P(x)P(y)$ (i.e., $\underline{x}$ and $\underline{y}$ are uncorrelated).

Furthermore, MI achieves its upper bound given by Eq.(93.9) if either

- $\underline{y}$ is a deterministic function of $\underline{x}$; i.e.,

$$P(y \mid x) = \mathbb{1}(\underline{y} = f(\underline{x})) \quad (93.10)$$

In that case, $\ln P(y \mid x)$ is either $0 \ln(0)$ or $\ln(1)$ so

$$H(\underline{y} \mid \underline{x}) = 0 \quad (93.11)$$

Furthermore,

$$H(\underline{y} \mid \underline{x}) = 0 \implies H(\underline{x} : \underline{y}) = H(\underline{y}) \quad (93.12)$$

- $\underline{x}$ is a deterministic function of $\underline{y}$; i.e.,

$$P(x \mid y) = \mathbb{1}(\underline{x} = g(\underline{y})) \quad (93.13)$$

In that case, $\ln P(x \mid y)$ is either $0 \ln(0)$ or $\ln(1)$ so

$$H(\underline{x} \mid \underline{y}) = 0 \quad (93.14)$$

Furthermore

$$H(\underline{x} \mid \underline{y}) = 0 \implies H(\underline{x} : \underline{y}) = H(\underline{x}) \quad (93.15)$$

Define the **MI Efficiency** $\epsilon(\underline{y} \mid \underline{x})$ by

$$\epsilon(\underline{y} \mid \underline{x}) = \frac{H(\underline{y} : \underline{x})}{H(\underline{y})} = 1 - \frac{H(\underline{y} \mid \underline{x})}{H(\underline{y})} \quad (93.16)$$

From $0 \leq H(\underline{y} : \underline{x}) \leq H(\underline{y})$ we get

$$0 \leq \epsilon(\underline{y} \mid \underline{x}) \leq 1 \quad (93.17)$$

From its definition, we see that the MI efficiency has a singularity of the type $0/0$ when $H(\underline{y} \mid \underline{x}) = H(\underline{y}) = 0$.

We will henceforth refer to the $(H(\underline{y}), H(\underline{y} \mid \underline{x}))$ plane as the **entropy versus conditional info (E-CI) plane**

It is instructive (see Fig.93.1) to plot in the E-CI plane, the loci on which the efficiency equals certain values.

Another thing that I find instructive is to calculate the values of the efficiency in the case that $\underline{x}$ and $\underline{y}$ are simple binary random variables $\underline{x}, \underline{y} \in \{0, 1\}$. In that case, we represent $P(y|x)$ by a 2×2 matrix, with $\underline{y} \in \{0, 1\}$ labelling the rows and $\underline{x} \in \{0, 1\}$ labelling the columns. Fig.93.2) gives the values of the efficiency for various $P(y|x)$ and $P(x)$ possibilities.

93.1.2 Standard Ising Model

The standard Ising model is defined on a cubic lattice, as shown in Fig.93.3.

Let

$$Bool = \{0, 1\}$$

$$S_i \in \{-1, +1\} = 2Bool - 1 = \mathbb{Z}_{\pm}$$

T = temperature

$\beta = 1/(k_B T)$. We will set Boltzmann constant k_B to 1.

D = dimension of lattice. We refer to $D = 1$ as 1-dim, and to $D = 2$ as 2-dim.

$NN(i)$ = set of indices of nearest neighbors to S_i .

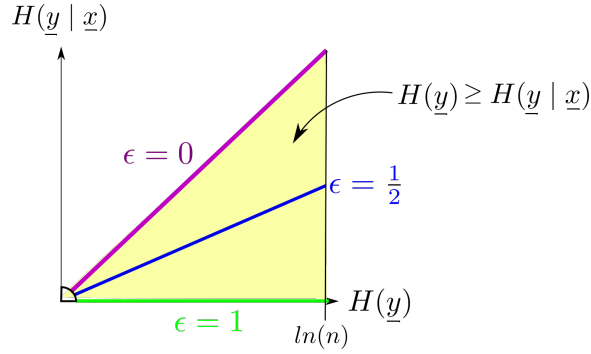


Figure 93.1: Efficiency $\epsilon = \frac{H(y|x)}{H(y)} = 1 - \frac{H(y|x)}{H(y)}$. Efficiency has a singularity when both $H(y|x)$ and $H(y)$ are zero.

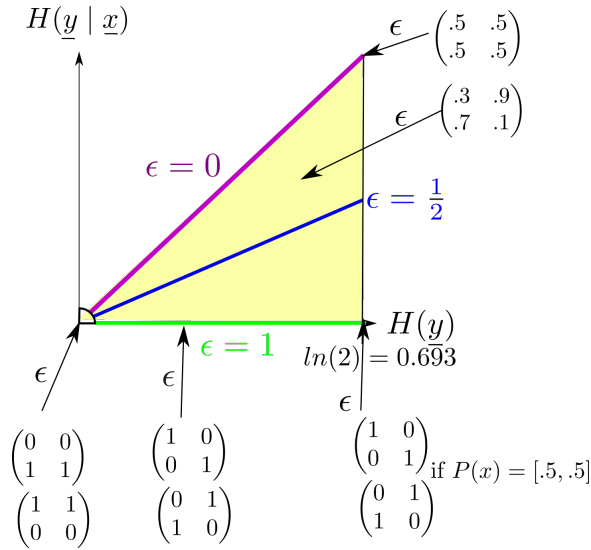


Figure 93.2: Efficiency for various values of the transition matrix $P(y|x)$ and the prior probabilities $P(x)$.

$z = 2D =$ **coordination number**, number of nearest neighbors internally in the lattice, away from the boundary. $z = 2$ in $D = 1$ and $z = 4$ in $D = 2$. For spins on the boundary, the number of NN is less than z . For example, as you can see in Fig.93.3, for a rectangular lattice in 2-dim, the corner sites have 2 NN, and the non-corner boundary sites have 3 NN. Only the internal sites have 4 NN.

Assume coupling constants $J, h \geq 0$.

The **D-dim Ising Model** is defined as follows.

For $i = 1, 2, 3, \dots, n$, define **energies** $\mathcal{E}_i$ and $\mathcal{E}$ by

$$\mathcal{E}_i = \left[-JS_i \sum_{j \in NN(i)} S_j \right] - hS_i, \quad \mathcal{E} = \sum_{i=1}^n \mathcal{E}_i \quad (93.18)$$

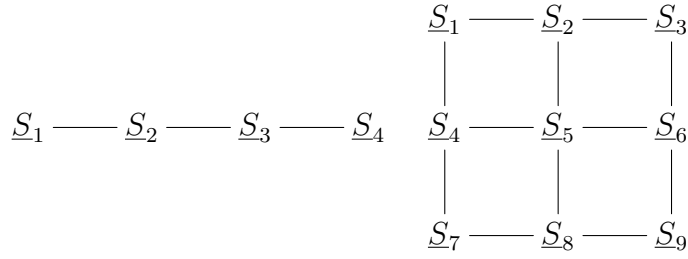


Figure 93.3: Ising Model cubic lattice for $D = 1$ (with 4 spins) and $D = 2$ (with 9 spins).

and a Partition Function Z by

$$Z = \left[\prod_{i=1}^n \sum_{S_i \in \mathbb{Z}_{\pm}} \right] e^{-\beta \mathcal{E}} \quad (93.19)$$

Let

$\xi =$ **correlation length**. $\langle \underline{S}_i, \underline{S}_{i+r} \rangle \sim e^{-r/\xi}$. $\xi \rightarrow \infty$ at critical points

$\langle m \rangle = \frac{1}{n} \sum_{i=1}^n \underline{S}_i =$ **magnetization**

$T_c = 1/\beta_c =$ **Curie Temperature** (a.k.a. critical temperature). Magnetization is non zero for $T < T_c$ (spontaneous magnetization) and zero for $T > T_c$ (disordered phase).

The 2-dim Ising model with $h = 0$ has a critical point at $T = T_c$. If either $D = 1$ or $h \neq 0$, there are no critical points at any finite temperature, just phase transitions.

Ising Model can only be solved exactly for $D = 1, 2$. For other D , must solve numerically.

For the 2-dim Ising model with $h = 0$:

- Curie Temperature

$$T_c = \frac{2J}{\ln(1 + \sqrt{2})} \quad (93.20)$$

$$\beta_c J = \frac{1}{2} \ln(1 + \sqrt{2}) \approx 0.4407 \quad (93.21)$$

- correlation length

$$\xi \sim |T - T_c|^{-\nu}, \quad \nu = 1 \quad (93.22)$$

93.2 Scaling a General Bnet

Consider a general bnet with nodes $\underline{x}_i$ for $i = 1, 2, \dots, n$. Let

$\mathcal{R}$ = set of root nodes.

$\mathcal{R}^c$ = set of non-root nodes.

$x^{:n} = (x_1, x_2, \dots, x^n)$

Then

$$P(x^{:n}) = \left[\prod_{\underline{x}_i \in \mathcal{R}^c} P(x_i \mid pa(\underline{x}_i)) \right] \prod_{\underline{x}_i \in \mathcal{R}} P(x_i) \quad (93.23)$$

For $1 < b < \infty$, let

$$P_b(x_i \mid pa(\underline{x}_i)) = \frac{P(x_i \mid pa(\underline{x}_i))^b}{Z_i(b)} = \pi_i(b) \quad (93.24)$$

where

$$Z_i(b) = \sum_{x_i \in \text{val}(x_i)} P(x_i \mid pa(\underline{x}_i))^b \quad (93.25)$$

As $b \rightarrow \infty$,

$(\pi_1, \pi_2, \dots, \pi_n)$ tends to a one hot vector; i.e., all its components tend to zero except for one component, the one that is smallest at $b = 1$, which tends to 1:

$$\pi_i(b) \rightarrow \delta(i, i_0), \quad \text{where } i_0 = \underset{i}{\operatorname{argmin}} \pi_i(b = 1) \quad (93.26)$$

Therefore, $H(\underline{x}_i \mid pa(\underline{x}_i))$ (for P_b) tends to a sum of terms of the form $0 \ln(0)$ or $\ln(1)$. Thus

$$H(\underline{x}_i \mid pa(\underline{x}_i)) \rightarrow 0 \quad (93.27)$$

Thus

$$H(\underline{x}_i : pa(\underline{x}_i)) \rightarrow H(\underline{x}_i) \quad (93.28)$$

If $H(\underline{x}_i)$ remains finite as $b \rightarrow \infty$, then $\epsilon(\underline{x}_i \mid pa(\underline{x}_i)) \rightarrow 1$, but if $H(\underline{x}_i)$ tends to zero, then $\epsilon(\underline{x}_i \mid pa(\underline{x}_i))$ tends to a $0/0$ singularity at $H(\underline{y} \mid \underline{x}) = H(\underline{y}) = 0$

Claim 140

$$\sum_{i=1}^n H(\underline{x}_i) - H(\underline{x}^{:n}) = \sum_{\underline{x}_i \in \mathcal{R}^c} H(\underline{x}_i : pa(\underline{x}_i)) \quad (93.29)$$

Note that both sides equal zero when all the nodes are uncorrelated.

proof:

$$H(\underline{x}_i : pa(\underline{x}_i)) = \sum_{x_i} \sum_{pa(\underline{x}_i)} P(x_i, pa(\underline{x}_i)) \ln \frac{P(x_i | pa(\underline{x}_i))}{P(x_i)} \quad (93.30)$$

$$H(\underline{x}_i) = - \sum_{x_i} P(x_i) \ln P(x_i) \quad (93.31)$$

$$H(\underline{x}^{\cdot n}) = - \sum_{x^{\cdot n}} P(x^{\cdot n}) \ln P(x^{\cdot n}) \quad (93.32)$$

From Eq.(93.23), we get

$$\frac{P(x^{\cdot n})}{\prod_{i=1}^n P(x_i)} = \prod_{\underline{x}_i \in \mathcal{R}^c} \frac{P(x_i | pa(\underline{x}_i))}{P(x_i)} \quad (93.33)$$

Hence

$$\sum_{x^{\cdot n}} P(x^{\cdot n}) \ln \frac{P(x^{\cdot n})}{\prod_{i=1}^n P(x_i)} = \sum_{\underline{x}_i \in \mathcal{R}^c} H(\underline{x}_i : pa(\underline{x}_i)) \quad (93.34)$$

QED

This last claim says that the “bound entropy” (i.e., the entropy between when n random variables are independent and when they are correlated) equals the sum of the mutual informations for the non-root nodes.

Claim 141

$$\max_i H(\underline{x} : \underline{a}_i) \leq H(\underline{x} : \underline{a}_1, \underline{a}_2, \dots, \underline{a}_m) \leq \sum_{i=1}^m H(\underline{x} : \underline{a}_i) \quad (93.35)$$

proof:

We will prove this for $m = 3$ and leave the general case to the reader.

Averaging the logarithm of the following chain rule for conditional probabilities

$$\frac{P(x, a_1, a_2, a_3)}{P(x)P(a_1, a_2, a_3)} = \left[\frac{P(x, a_1 | a_2, a_3)}{P(x | a_2, a_3)P(a_1 | a_2, a_3)} \right] \left[\frac{P(x, a_2 | a_3)}{P(x | a_3)P(a_2 | a_3)} \right] \frac{P(x, a_3)}{P(x)P(a_3)} \quad (93.36)$$

we get

$$H(\underline{x} : \underline{a}_1, \underline{a}_2, \underline{a}_3) = H(\underline{x} : \underline{a}_1 | \underline{a}_2, \underline{a}_3) + H(\underline{x} : \underline{a}_2 | \underline{a}_3) + H(\underline{x} : \underline{a}_3) \quad (93.37)$$

Both sides of the inequality Eq.(93.35) follow from Eq.(93.37). Indeed,

$$H(\underline{x} : \underline{a}_1, \underline{a}_2, \underline{a}_3) = \underbrace{H(\underline{x} : \underline{a}_1 | \underline{a}_2, \underline{a}_3)}_{\leq H(\underline{x} : \underline{a}_1)} + \underbrace{H(\underline{x} : \underline{a}_2 | \underline{a}_3)}_{\leq H(\underline{x} : \underline{a}_2)} + H(\underline{x} : \underline{a}_3) \quad (93.38)$$

$\underbrace{\hspace{10em}}_{\geq H(\underline{x} : \underline{a}_3)}$

QED

Dividing Eq.(93.35) by $H(\underline{x})$ and identifying the $\underline{a}_i$ nodes with the parents $pa(\underline{x})$ of $\underline{x}$, we get

$$\max_{\underline{a} \in pa(\underline{x})} \epsilon(\underline{x}|\underline{a}) \leq \epsilon(\underline{x} \mid pa(\underline{x})) \leq \sum_{\underline{a} \in pa(\underline{x})} \epsilon(\underline{x} \mid \underline{a}) \quad (93.39)$$

93.3 Scaling our Ising-dbnet Model

So far, we have described a method of scaling any bnet. In this section, we define an Ising-dbnet and apply our scaling method to it. Our Ising-dbnet is designed to incorporate all the physics of the standard Ising model.

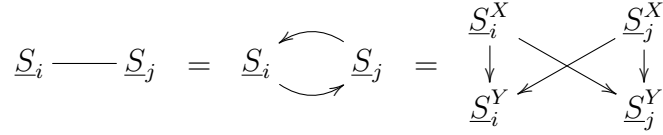


Figure 93.4: DAG definition of undirected arrow.

Let

$Bool = \{0, 1\}$

$S_i, S_i^X, S_i^Y \in \{1, -1\} = 2Bool - 1 = \mathbb{Z}_\pm$

$D_i = (S_i^X, S_i^Y) =$ “dipole node” (dnode) or “lattice site”

$T =$ temperature

$b = \beta = 1/T$

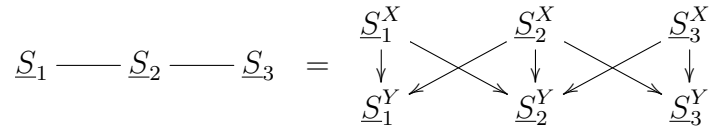


Figure 93.5: DAG definition of 1-dim Ising Model with 3 dnodes.

Assume coupling constants $J, h, \lambda \geq 0$. Actually, in our software `bnet_rg` described later on, we only consider the case $h = \lambda = 0$.

93.3.1 1-dim Ising bnet Model

Our 1-dim Ising bnet model with 3 dnodes is defined as follows (generalization to n dnodes is obvious).

Consider the bnet of Fig.93.5 with node Transition Probability Matrices (TPM) given below, written in blue.

For $i = 1, 2, 3$,

$$P(S_i^X) = \frac{1}{2} \quad (93.40)$$

For $i = 2$,

$$\begin{cases} P(S_2^Y | S_2^X, S_1^X, S_3^X) = \frac{e^{-\beta \mathcal{E}_2}}{Z_2} \\ \mathcal{E}_2 = -JS_2^Y(S_1^X + S_3^X) - hS_2^Y - \lambda S_2^Y S_2^X \\ Z_2 = \sum_{S_2^Y \in \mathbb{Z}_\pm} e^{-\beta \mathcal{E}_2} \end{cases} \quad (93.41)$$

For $i = 1, 3$

$$\begin{cases} P(S_i^Y | S_i^X, S_2^X) = \frac{e^{-\beta \mathcal{E}_i}}{Z_i} \\ \mathcal{E}_i = -JS_i^Y(S_2^X) - hS_i^Y - \lambda S_i^Y S_i^X \\ Z_i = \sum_{S_i^Y \in \mathbb{Z}_\pm} e^{-\beta \mathcal{E}_i} \end{cases} \quad (93.42)$$

93.3.2 2-dim Ising bnet Model

Let

$\mathcal{B}$ = border dnodes of undirected graph

$\mathcal{B}^c$ = non-border, internal dnodes of undirected graph

n = number of nodes of undirected graph

Our 2-dim Ising bnet model with 9 dnodes is defined as follows (generalization to n dnodes is obvious).

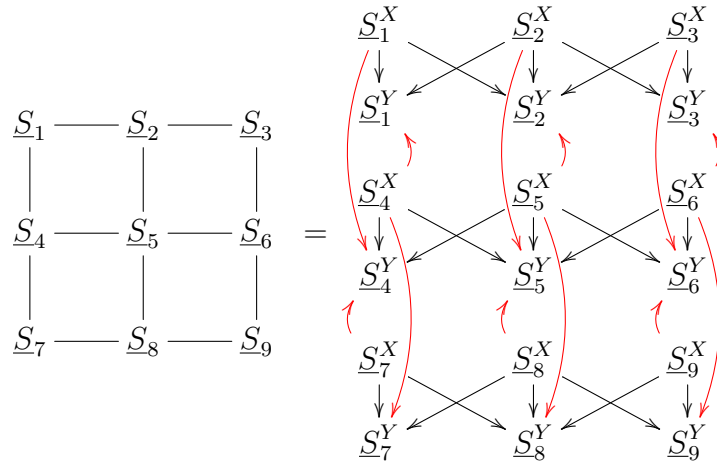


Figure 93.6: DAG definition of 2-dim Ising bnet with 9 dnodes on a 3×3 lattice.

Consider the bnet of Fig.93.6 with node Transition Probability Matrices (TPM) given below, written in blue.

For $i = 1, 2, \dots, n$

$$P(S_i^X) = \frac{1}{2} \quad (93.43)$$

For $\underline{S}_i \in \mathcal{B}^c$, if $\underline{S}_a, \underline{S}_b, \underline{S}_c, \underline{S}_d$ are the nearest neighbors (NN) of $\underline{S}_i$,

$$\begin{cases} P(S_i^Y | S_i^X, S_a^X, S_b^X, S_c^X, S_d^X) = \frac{e^{-\beta \mathcal{E}_i}}{Z_i} \\ \mathcal{E}_i = -JS_i^Y(S_i^X + S_a^X + S_b^X, S_c^X + S_d^X) - hS_i^Y - \lambda S_i^Y S_i^X \\ Z_i = \sum_{S_i^Y \in \mathbb{Z}_\pm} e^{-\beta \mathcal{E}_i} \end{cases} \quad (93.44)$$

For $\underline{S}_i \in \mathcal{B}$, the corner nodes have 2 NN and the non-corner ones have 3 NN. Eq.(93.44) is still valid for $\underline{S}_i \in \mathcal{B}$ if we set $\underline{S}_c = \underline{S}_d = 0$ in case of 2 NN and $\underline{S}_d = 0$ in case of 3 NN.

For the full bnet,

$$P(D^n) = \left[\prod_{i=1}^n P(S_i^Y | pa(\underline{S}_i^Y)) \right] \prod_{i=1}^n P(S_i^X) \quad (93.45)$$

For a 2-dim Ising bnet, Claim 140 can be specialized as follows. Using $D_i = (S_i^X, S_i^Y)$ and the fact that the S_i^X nodes are all root nodes.

$$\underbrace{\sum_{i=1}^n H(\underline{x}_i)}_{\sum_{i=1}^n H(D_i)} - \underbrace{H(\underline{x}^n)}_{H(\underline{D}^n)} = \underbrace{\sum_{\underline{x}_i \in \mathcal{R}^c} H(\underline{x}_i : pa(\underline{x}_i))}_{\sum_{i=1}^n H(\underline{S}_i^Y : pa(\underline{S}_i^Y))} \quad (93.46)$$

93.3.3 Converting 1-time bnet to a dynamical one

The 1-time Ising bnet presented so far does not equilibrate. To build a bnet that does equilibrate, we can make many copies (“times-slices”) of the above 1-time bnet and connect them to make a dynamic bnet.

Add a functional dependance (t) to all the random variables $\underline{D}_i$ in the 1-time bnet presented above. Do the same with $(t+1)$. Now we have two subsequent time slices.

To connect them, set $S_i^X(t+1)$ equal to $S_i^Y(t)$ for all i . In other words, we need to add the following TPM

$$P(S_i^X(t+1) | S_i^Y(t)) = \mathbb{1}(S_i^X(t+1) = S_i^Y(t)) \quad (93.47)$$

for all i .

93.4 Software

The ideas in this paper are implemented as free open source Python software at the github repo `bnet_rg` (see Ref. [94])

Let

$\hat{\beta} = \beta/\beta_c$, where $\beta_c = 1/T_c$. $\hat{\beta} = 1$ at the Curie temperature T_c for a 2-dim Ising model.

In all our Jupyter notebooks, we take $J = 1$ and $h = \lambda = 0$

We consider a lattice that is 5×5 . At each lattice site $i = 1, 2, \dots, 25$, there are two nodes S_i^X and S_i^Y .

All the S_i^X nodes are root nodes of time slice t . The $S_a^X, S_b^X, S_c^X, S_d^X$ nodes of the nearest neighbors of site i point to S_i^Y

Iteration carries us from time slice t to time slice $t + 1$. At the beginning of iteration $t + 1$, we set $S_i^X(t + 1)$ equal to $S_i^Y(t)$ for all i .

p_0 is the initial ($t = 1$) probability $P(S_i^X(t = 1) = -1)$ for all i . A separate Jupyter notebook is provided for $p_0=0.3, 0.5, 0.7, p_0=None$, etc. ($p_0=None$ means each node is assigned a different p_0 sampled from the $\text{uniform}([0,1])$ probability distribution.

We present 3 types of plots

1. **lattice plots** (see Fig.93.7)

These show a 5×5 lattice with a color bar for efficiency ranging from 0 to 1.

Each of the 25 lattice sites has 2 arrows connecting it to each of its nearest neighbors.

The arrows are each labelled with the value of the efficiency for that arrow.

Sometimes the efficiency of an arrow is undefined. In that case, the arrow is dashed.

We use the probabilities $P(S_i^Y = \pm 1)$ to sample each node. If the sample turns out to be -1 (resp., $+1$), the node appears black (resp. white).

2. **parametric plots** (see Fig.93.8)

These show a parametric curve in the (x, y) plane, where

$$x = \text{average entropy} = \frac{1}{n} \sum_{i=1}^n H(S_i^Y)$$

$$y = \text{average conditional information} = \frac{1}{n} \sum_{i=1}^n H(S_i^Y \mid S_a^X, S_b^X, S_c^X, S_d^X)$$

$$n = 25$$

The parameter of the curves is $\hat{\beta}$.

The first and last point of the parametric curve are labelled by the value of the parameter $\hat{\beta}$.

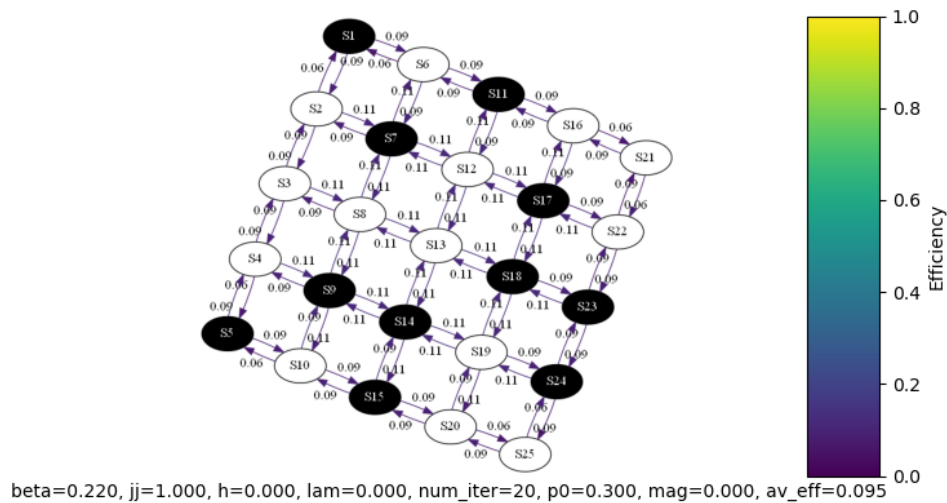


Figure 93.7: Example of lattice plot.

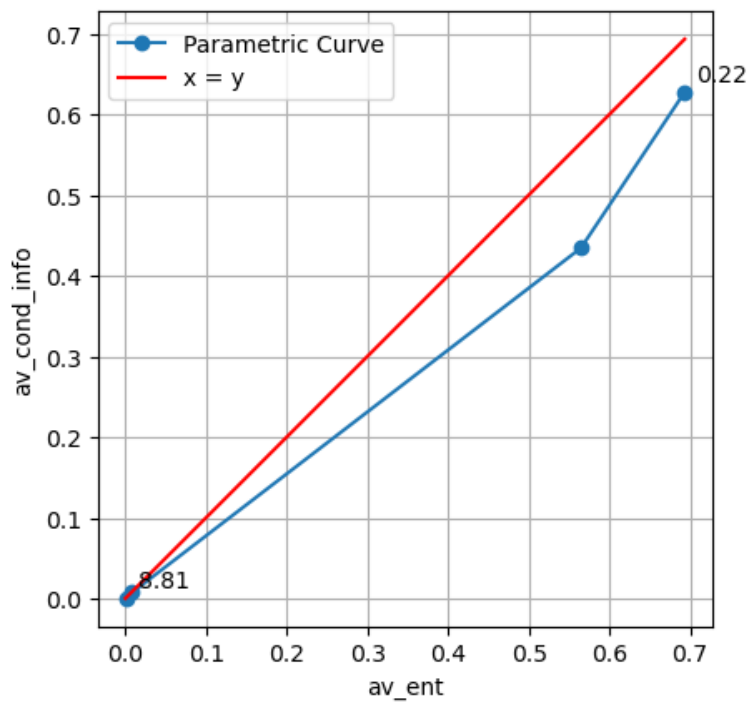


Figure 93.8: Example of parametric plot.

3. Magnetization plots(see Fig.93.9)

These show the magnetization $\text{mag} = \frac{1}{n} \sum_{i=1}^n S_i^Y$ as a function of $\text{beta_hat} = \hat{\beta}$

Conclusions

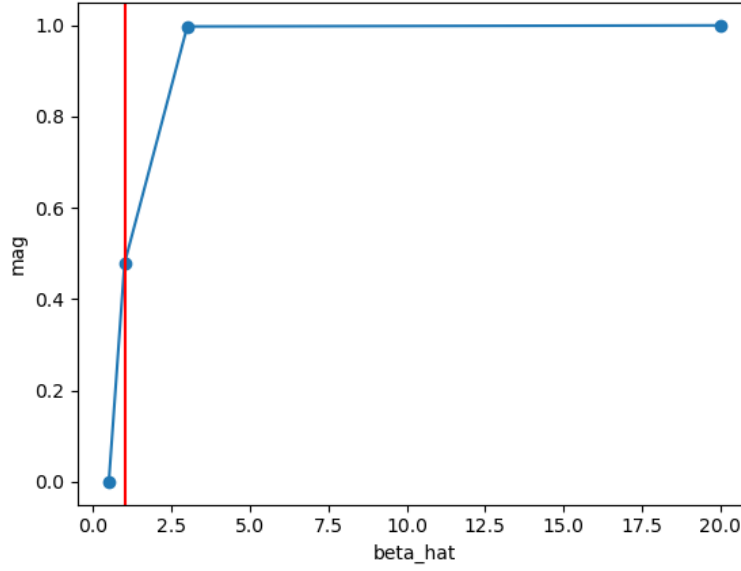


Figure 93.9: Example of magnetization plot.

1. Our 2-dim Ising-dbnet model exhibits a discontinuity in the magnetization at the same Curie temperature as the standard vanilla 2-dim Ising model.
2. For $p_0 \neq .5$, the parametric curves flow smoothly as $\beta \rightarrow \infty$, towards $(H(y), H(y | x)) = (0, 0)$, where the efficiency is discontinuous.
4. For $p_0 = .5$, the parametric curves flow smoothly as $\beta \rightarrow \infty$, vertically down.

Bibliography

- [1] Alan Agresti. *An introduction to categorical data analysis*. John Wiley & Sons, 2018.
- [2] Agrumery. PyArum, free open source app at github. <https://github.com/agrumery/aGrUM>.
- [3] Uri Alon. *An introduction to systems biology: design principles of biological circuits*. Chapman and Hall/CRC, 2019.
- [4] Data Analytics and IIT Delhi Intelligence Research (DAIR) Group. Openie6. <https://github.com/dair-iitd/openie6>.
- [5] Elias Bareinboim, Jin Tian, and Judea Pearl. Recovering from selection bias in causal and statistical inference. In *Twenty-Eighth AAAI Conference on Artificial Intelligence*, 2014. https://ftp.cs.ucla.edu/pub/stat_ser/r425.pdf.
- [6] Dan Bendel. Metropolis-Hastings: A comprehensive overview and proof. <https://similarweb.engineering/mcmc/>.
- [7] David Benkeser and Antoine Chambaz. A ride in targeted learning territory. <https://achambaz.github.io/tlride/tlride-book.pdf>.
- [8] Brilliant.org. Finite state machines. <https://brilliant.org/wiki/finite-state-machines/>.
- [9] Rafael Cabañas de Paz, Alessandro Antonucci, Andrés Cano Utrera, Manuel Gómez-Olmedo, et al. Evaluating interval-valued influence diagrams. 2017. <https://digibug.ugr.es/bitstream/handle/10481/67901/antonucci2016b.pdf?sequence=1>.
- [10] Alexandra M Carvalho. Scoring functions for learning Bayesian networks. http://www.lx.it.pt/~asmc/pub/talks/09-TA/ta_pres.pdf.
- [11] Bo Chang. Copula: a very short introduction, article in Bo’s Blog. <https://bochang.me/blog/posts/copula/>.
- [12] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. <https://arxiv.org/abs/1603.02754>.

- [13] Victor Chernozhukov, Carlos Cinelli, Whitney Newey, Amit Sharma, and Vasilis Syrgkanis. Long story short: Omitted variable bias in causal machine learning. https://www.nber.org/system/files/working_papers/w30302/w30302.pdf.
- [14] Carlos Cinelli, Andrew Forney, and Judea Pearl. A crash course in good and bad controls. https://ftp.cs.ucla.edu/pub/stat_ser/r493.pdf.
- [15] Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. [https://carloscinelli.com/files/Cinelli%20and%20Hazlett%20\(2020\)%20-%20Making%20Sense%20of%20Sensitivity.pdf](https://carloscinelli.com/files/Cinelli%20and%20Hazlett%20(2020)%20-%20Making%20Sense%20of%20Sensitivity.pdf).
- [16] Scott Cunningham. *Causal inference: The mixtape*. Yale University Press, 2021. <https://mixtape.scunning.com/index.html>.
- [17] Nir Friedman Daphne Koller. *Probabilistic Graphical Models, Principles and Techniques*. MIT Press, 2009.
- [18] Robin J. Evans. Graphical methods for inequality constraints in marginalized DAGs. <https://arxiv.org/abs/1209.2978>.
- [19] Matheus Facure Alves. *Causal Inference for The Brave and True*. 2021. <https://matheusfacure.github.io/python-causality-handbook/landing-page.html>.
- [20] Richard P Feynman, Albert R Hibbs, and Daniel F Styer. *Quantum mechanics and path integrals*. Courier Corporation, 2010.
- [21] Charles Fox, Neil Girdhar, and Kevin Gurney. A causal Bayesian network view of reinforcement learning. <https://www.aaai.org/Papers/FLAIRS/2008/FLAIRS08-030.pdf>.
- [22] Armin Fuchs. Dynamical systems in one and two dimensions: A geometrical approach. In *Nonlinear Dynamics in Human Behavior*, pages 1–33. Springer, 2011.
- [23] Nigel Goldenfeld. *Lectures on phase transitions and the renormalization group*. CRC Press, 2018.
- [24] Bruno Gonçalves. Model testing and causal search. blog post <https://medium.com/data-for-science/causal-inference-part-vii-model-testing-and-causal-search-536b796f0384>.
- [25] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, David Warde-Farley Bing Xu, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks. <https://arxiv.org/abs/1406.2661>.

- [26] B.S. Jeffreys H. Jeffreys. *Methods of Mathematical Physics, 3rd ed.* Cambridge University Press, 1988.
- [27] James Douglas Hamilton. *Time series analysis.* Princeton University Press, 2020.
- [28] Sebastian Haneuse and Andrea Rotnitzky. Estimation of the effect of interventions that modify the received treatment. *Statistics in medicine*, 32(30):5260–5277, 2013. Main:<https://sci-hub.se/10.1002/sim.5907>, Supplement: <https://onlinelibrary.wiley.com/action/downloadSupplement?doi=10.1002%2Fsim.5907&file=sim5907-sup-0001-Appendix.pdf>.
- [29] Christina Heinze-Deml. Causality, spring semester 2019 at ETH Zurich. https://stat.ethz.ch/lectures/ss19/causality.php#course_materials.
- [30] MA Hernán and J Robins. Causal inference: What if. Boca Raton: Chapman & Hill/CRC, <https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/>.
- [31] Katherine Hoffman. An illustrated guide to modified treatment policies, part 1: Introduction and motivation, article in KHstats blog. <https://www.khstats.com/blog/lmtp/lmtp/>.
- [32] Katherine Hoffman. An illustrated guide to TMLE, article in KHstats blog. <https://www.khstats.com/blog/tmle/tutorial/>.
- [33] Cecil Huang and Adnan Darwiche. Inference in belief networks: A procedural guide. *International journal of approximate reasoning*, 15(3):225–263, 1996. <http://www.ar-tiste.com/Huang-Darwiche1996.pdf>.
- [34] Janos Körner Imre Csiszár. *Information Theory- Coding Theorems for Discrete Memoryless Systems.* Academic Press, 1981.
- [35] Tommi S. Jaakkola and Michael I. Jordan. Variational probabilistic inference and the QMR-DT network. <http://arxiv.org/abs/1105.5462>.
- [36] Michael I. Jordan. course: Stat260-Bayesian modeling and inference, lecture date: February 8-2010, title: The conjugate prior for the Normal distribution. <https://people.eecs.berkeley.edu/~jordan/courses/260-spring10/lectures/lecture5.pdf>.
- [37] Chaitanya K. Joshi. Transformer (machine learning model). <https://graphdeeplearning.github.io/post/transformers-are-gnns/>.
- [38] Keshav Kolluru, Vaibhav Adlakha, Samarth Aggarwal, Mausam, and Soumen Chakrabarti. Openie6: Iterative grid labeling and coordination analysis for open information extraction. <https://arxiv.org/abs/2010.03147>.

- [39] Chung-Ming Kuan. Introduction to time series analysis, Fall 2014 lectures given at the Department of Finance, National Taiwan University. https://homepage.ntu.edu.tw/~ckuan/pdf/2014fall/Lec-TimeSeries_slide-Fall2014.pdf.
- [40] Steffen L Lauritzen and Dennis Nilsson. Representing and solving decision problems with limited information. *Management Science*, 47(9):1235–1251, 2001. https://people.duke.edu/~rnau/DAS_pub_award_2003/lauritzen_nilsson.pdf.
- [41] Steffen L Lauritzen and David J Spiegelhalter. Local computations with probabilities on graphical structures and their application to expert systems. *Journal of the Royal Statistical Society: Series B (Methodological)*, 50(2):157–194, 1988. <http://www.eecis.udel.edu/~shatkay/Course/papers/Lauritzen1988.pdf>.
- [42] Boram Lee. Lecture notes by Boram Lee, September 28, 2015.
- [43] Sergey Levine. Course CS 285 at UC Berkeley, Deep reinforcement learning. <http://rail.eecs.berkeley.edu/deeprlcourse/>.
- [44] Ang Li. Ph.D. Thesis, UCLA 2021. https://ftp.cs.ucla.edu/pub/stat_ser/r507.pdf.
- [45] Juliane Liepe, Sarah Filippi, Michał Komorowski, and Michael PH Stumpf. Maximizing the information content of experiments in systems biology. *PLoS computational biology*, 9(1):e1002888, 2013. <https://journals.plos.org/ploscompbiol/article/file?id=10.1371/journal.pcbi.1002888&type=printable>.
- [46] Dimitris Margaritis. Learning Bayesian network model structure from data (thesis, 2003, Carnegie Mellon Univ). <https://apps.dtic.mil/sti/citations/ADA461103>.
- [47] Adam A Margolin, Ilya Nemenman, Katia Basso, Chris Wiggins, Gustavo Stolovitzky, Riccardo Dalla Favera, and Andrea Califano. Aracne: an algorithm for the reconstruction of gene regulatory networks in a mammalian cellular context. In *BMC bioinformatics*, volume 7, page S7. Springer, 2006. <https://link.springer.com/article/10.1186/1471-2105-7-S1-S7>.
- [48] Samuele Mazzanti. Black-box models are actually more explainable than a logistic regression. <https://towardsdatascience.com/black-box-models-are-actually-more-explainable-than-a-logistic-regression-f263c22795d>.

- [49] Samuele Mazzanti. SHAP values explained exactly how you wished someone explained to you. <https://towardsdatascience.com/shap-explained-the-way-i-wish-someone-explained-it-to-me-ab81cc69ef30>.
- [50] William D McComb. *Renormalization methods: a guide for beginners*. Oxford University Press, 2004.
- [51] Robert J. McEliece, David J. C. MacKay, and Jung-Fu Cheng. Turbo decoding as an instance of Pearl’s belief propagation algorithm. <http://authors.library.caltech.edu/6938/1/MCEieeejstc98.pdf>.
- [52] Scott Mueller, Ang Li, and Judea Pearl. Causes of effects: Learning individual responses from population data. *arXiv preprint arXiv:2104.13730*, 2021. <https://arxiv.org/abs/2104.13730>.
- [53] Brady Neal. Introduction to causal inference, Fall 2020, lectures and book. <https://www.bradyneal.com/causal-inference-course>.
- [54] Richard E Neapolitan. *Learning Bayesian networks*. Pearson Prentice Hall, 2004.
- [55] Andrew Ng. Lecture at deeplearning.ai on recurrent neural networks. <http://www.ar-tiste.com/ng-lec-rnn.pdf>.
- [56] Alexandru Niculescu-Mizil and Rich Caruana. Inductive transfer for bayesian network structure learning. In *Artificial intelligence and statistics*, pages 339–346. PMLR, 2007. <http://proceedings.mlr.press/v2/niculescu-mizil07a/niculescu-mizil07a.pdf>.
- [57] Gregory Nuel. Tutorial on exact belief propagation in Bayesian networks: from messages to algorithms. <https://arxiv.org/abs/1201.4724>.
- [58] paperspace.com. PyTorch 101, Part 1: Understanding graphs, automatic differentiation and autograd. <https://blog.paperspace.com/pytorch-101-understanding-graphs-and-automatic-differentiation/>.
- [59] Kalyanapuram R Parthasarathy. *An introduction to quantum stochastic calculus*, volume 85. Birkhäuser, 2012.
- [60] Judea Pearl. Linear models: A useful microscope for causal analysis. https://ftp.cs.ucla.edu/pub/stat_ser/r409-corrected-reprint.pdf.
- [61] Judea Pearl. Mediating instrumental variables. https://ftp.cs.ucla.edu/pub/stat_ser/r210.pdf.
- [62] Judea Pearl. On the testability of causal models with latent and instrumental variables. <https://arxiv.org/abs/1302.4976>.

- [63] Judea Pearl. Reverend Bayes on inference engines: A distributed hierarchical approach. <https://www.aaai.org/Papers/AAAI/1982/AAAI82-032.pdf>, 1982.
- [64] Judea Pearl. *Probabilistic Inference in Intelligent Systems*. Morgan Kaufmann, 1988.
- [65] Judea Pearl. The causal mediation formula—a guide to the assessment of pathways and mechanisms. *Prevention science*, 13(4):426–436, 2012. <https://apps.dtic.mil/sti/pdfs/ADA557663.pdf>.
- [66] Judea Pearl. *Causality: Models, Reasoning, and Inference, Second Edition*. Cambridge University Press, 2013.
- [67] Judea Pearl. Causal and counterfactual inference. *The Handbook of Rationality*, pages 1–41, 2019. https://ftp.cs.ucla.edu/pub/stat_ser/r485.pdf.
- [68] Judea Pearl and Elias Bareinboim. Transportability of causal and statistical relations: A formal approach. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 25, 2011. <https://ojs.aaai.org/index.php/AAAI/article/view/7861>.
- [69] Judea Pearl, Madelyn Glymour, and Nicholas P Jewell. *Causal inference in statistics: A primer*. John Wiley & Sons, 2016.
- [70] Judea Pearl and Dana Mackenzie. *The book of why: the new science of cause and effect*. Basic Books, 2018.
- [71] Judea Pearl and James M Robins. Probabilistic evaluation of sequential plans from causal models with hidden variables. *arXiv preprint arXiv:1302.4977*, 2013. <https://arxiv.org/abs/1302.4977>.
- [72] Ashwin Rao and Tikhon Jelvis. *Foundations of Reinforcement Learning with Applications in Finance*. <https://stanford.edu/~ashlearn/RLForFinanceBook/book.pdf>.
- [73] ReliaSoft. System analysis reference. http://reliawiki.org/index.php/System_Analysis_Reference.
- [74] Walter Rudin. *Principles of Mathematical Analysis, 3rd ed*. McGraw-Hill, 1976.
- [75] Simo Särkkä and Arno Solin. *Applied stochastic differential equations*, volume 10. Cambridge University Press, 2019. https://users.aalto.fi/~ssarkka/pub/sde_book.pdf.

- [76] Simo Särkkä and Arno Solin. *Applied stochastic differential equations*. Cambridge University Press, 2019. https://users.aalto.fi/~ssarkka/pub/sde_book.pdf.
- [77] Scholarpedia. Granger causality. http://www.scholarpedia.org/article/Granger_causality.
- [78] Marco Scutari. bnlearn. <https://www.bnlearn.com/>.
- [79] Marco Scutari, Catharina Elisabeth Graafland, and José Manuel Gutiérrez. Who learns better Bayesian network structures: Accuracy and speed of structure learning algorithms. *International Journal of Approximate Reasoning*, 115:235–253, 2019. <https://arxiv.org/abs/1805.11908>.
- [80] Ross D Shachter and Debarun Bhattacharjya. Solving influence diagrams: Exact algorithms. *Wiley encyclopedia of operations research and management science.*, 2010. <http://www.ar-tiste.com/Shachter2010.pdf>.
- [81] Macksim Shevchenko. scikit-uplift. <https://github.com/maks-sh/scikit-uplift>.
- [82] Mark Srednicki. *Quantum field theory*. Cambridge university press, 2007.
- [83] Nitish Srivastava, G E Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan R Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. <http://www.cs.toronto.edu/~rsalakhu/papers/srivastava14a.pdf>.
- [84] StatQuest. XGBoost Parts 1-4 (videos). <https://statquest.org/video-index/>.
- [85] Russell Stuart and Norvig Peter. Artificial intelligence-a modern approach 3rd ed, 2016.
- [86] Masayoshi Takahashi. Statistical inference in missing data by MCMC and non-MCMC multiple imputation algorithms: Assessing the effects of between-imputation iterations. *Data Science Journal*, 16, 2017. <https://datascience.codata.org/articles/10.5334/dsj-2017-037/>.
- [87] theinvestorsbook.com. Pert analysis. <https://theinvestorsbook.com/pert-analysis.html>.
- [88] Joy A. Thomas Thomas M. Cover. *Elements of Information Theory*. Wiley-Interscience, 1991.
- [89] Jin Tian and Judea Pearl. Probabilities of causation: Bounds and identification. *Annals of Mathematics and Artificial Intelligence*, 28(1):287–313, 2000. https://ftp.cs.ucla.edu/pub/stat_ser/r271-A.pdf.

- [90] Michael E Tipping and Christopher M Bishop. Probabilistic principal component analysis. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 61(3):611–622, 1999. https://publications.aston.ac.uk/id/eprint/38367/1/NCRG_97_010.pdf.
- [91] Robert R. Tucci. Bayes-Petri net. https://github.com/rrtucci/Bayes_Petri_Net.
- [92] Robert R. Tucci. Bayesian networks (a.k.a. causal models, DAGs) and the passage of time. blog post in blog Quantum Bayesian Networks, <https://qbnets.wordpress.com/2021/07/16/bayesian-networks-aka-causal-models-dags-and-the-passage-of-time/>.
- [93] Robert R. Tucci. Bell’s inequalities for Bayesian statisticians. blog post in blog Quantum Bayesian Networks, <https://qbnets.wordpress.com/2008/09/19/bells-inequaties-for-bayesian-statistician/>.
- [94] Robert R. Tucci. bnet_rg. https://github.com/rrtucci/bnet_rg.
- [95] Robert R. Tucci. CausalFitbit. <https://github.com/rrtucci/CausalFitbit>.
- [96] Robert R. Tucci. Dag Lie Detector. https://github.com/rrtucci/DAG_Lie_Detector.
- [97] Robert R. Tucci. Goodness of causal fit. <https://github.com/rrtucci/goodness-c-fit>.
- [98] Robert R. Tucci. JudeasRx. <https://github.com/rrtucci/JudeasRx>.
- [99] Robert R. Tucci. Mappa Mundi. https://github.com/rrtucci/mappa_mundi.
- [100] Robert R. Tucci. Ode to odes (OTO). https://github.com/rrtucci/ode_to_odes. If a bnet in a figure in this book has the symbols ✓OTO next to it, then it is analyzed in OTO.
- [101] Robert R. Tucci. Quantum d-separation and quantum belief propagation. <https://arxiv.org/abs/2012.09635>.
- [102] Robert R. Tucci. Quantum Fog. <https://github.com/artiste-qb-net/quantum-fog>.
- [103] Robert R. Tucci. SCuMpy. <https://github.com/rrtucci/scumpy>.
- [104] Robert R. Tucci. SentenceAx. <https://github.com/rrtucci/SentenceAx>.
- [105] Robert R. Tucci. Shannon information theory without shedding tears over delta & epsilon proofs or typical sequences. <https://arxiv.org/abs/1208.2737>.

- [106] Robert R. Tucci. texnn. <https://github.com/rrtucci/texnn>.
- [107] Robert R. Tucci. Uplift Rocket. https://github.com/rrtucci/uplift_rocket.
- [108] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. <https://arxiv.org/abs/1706.03762>.
- [109] W. E. Vesely, F. F. Goldberg, N. H. Roberts, and D. F. Haasl. Fault tree handbook nureg-0492. <https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr0492/>.
- [110] Martin J Wainwright and Michael I Jordan. Graphical models, exponential families, and variational inference (book). https://people.eecs.berkeley.edu/~wainwrig/Papers/WaiJor08_FTML.pdf.
- [111] Lilian Weng. The multi-armed bandit problem and its solutions. lilianweng.github.io/lil-log, <http://lilianweng.github.io/lil-log/2018/01/23/the-multi-armed-bandit-problem-and-its-solutions.html>, 2018.
- [112] Lilian Weng. What are diffusion models? lilianweng.github.io/lil-log, <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>, 2021.
- [113] Wikibooks. Control systems. https://en.wikibooks.org/wiki/Control_Systems.
- [114] Wikipedia. AdaBoost. <https://en.wikipedia.org/wiki/AdaBoost>.
- [115] Wikipedia. Autoregressive model. https://en.wikipedia.org/wiki/Autoregressive_model.
- [116] Wikipedia. Autoregressive moving-average model. https://en.wikipedia.org/wiki/Autoregressive%E2%80%93moving-average_model.
- [117] Wikipedia. Baum-Welsh algorithm. https://en.wikipedia.org/wiki/Baum%E2%80%93Welch_algorithm.
- [118] Wikipedia. Belief propagation. https://en.wikipedia.org/wiki/Belief_propagation.
- [119] Wikipedia. Berkson’s paradox. https://en.wikipedia.org/wiki/Berkson%27s_paradox.
- [120] Wikipedia. Bernoulli distribution. https://en.wikipedia.org/wiki/Bernoulli_distribution.

- [121] Wikipedia. BERT language model. [https://en.wikipedia.org/wiki/BERT_\(language_model\)](https://en.wikipedia.org/wiki/BERT_(language_model)).
- [122] Wikipedia. Beta distribution. https://en.wikipedia.org/wiki/Beta_distribution.
- [123] Wikipedia. Beta function. https://en.wikipedia.org/wiki/Beta_function.
- [124] Wikipedia. Binary decision diagram. https://en.wikipedia.org/wiki/Binary_decision_diagram.
- [125] Wikipedia. Biplot. <https://en.wikipedia.org/wiki/Biplot>.
- [126] Wikipedia. Boolean algebra. https://en.wikipedia.org/wiki/Boolean_algebra.
- [127] Wikipedia. Bootstrap aggregating. https://en.wikipedia.org/wiki/Bootstrap_aggregating.
- [128] Wikipedia. Brownian dynamics. https://en.wikipedia.org/wiki/Brownian_dynamics.
- [129] Wikipedia. Categorical distribution. https://en.wikipedia.org/wiki/Categorical_distribution.
- [130] Wikipedia. Chi-square distribution. https://en.wikipedia.org/wiki/Chi-square_distribution.
- [131] Wikipedia. Chow-Liu tree. https://en.wikipedia.org/wiki/Chow%E2%80%93Liu_tree.
- [132] Wikipedia. Cochran's theorem. https://en.wikipedia.org/wiki/Cochran's_theorem.
- [133] Wikipedia. Conjugate prior. https://en.wikipedia.org/wiki/Conjugate_prior.
- [134] Wikipedia. Copula. [https://en.wikipedia.org/wiki/Copula_\(probability_theory\)](https://en.wikipedia.org/wiki/Copula_(probability_theory)).
- [135] Wikipedia. Cramer-Rao bound. https://en.wikipedia.org/wiki/Cram%C3%A9r%E2%80%93Rao_bound.
- [136] Wikipedia. Cross-validation. [https://en.wikipedia.org/wiki/Cross-validation_\(statistics\)](https://en.wikipedia.org/wiki/Cross-validation_(statistics)).
- [137] Wikipedia. Data processing inequality. https://en.wikipedia.org/wiki/Data_processing_inequality.

- [138] Wikipedia. Deterministic finite automaton. https://en.wikipedia.org/wiki/Deterministic_finite_automaton.
- [139] Wikipedia. Dimensional analysis. https://en.wikipedia.org/wiki/Dimensional_analysis.
- [140] Wikipedia. Dimensionality reduction. https://en.wikipedia.org/wiki/Dimensionality_reduction.
- [141] Wikipedia. Dirichlet distribution. https://en.wikipedia.org/wiki/Dirichlet_distribution.
- [142] Wikipedia. Errors in variables models. https://en.wikipedia.org/wiki/Errors-in-variables_models.
- [143] Wikipedia. Expectation maximization. https://en.wikipedia.org/wiki/Expectation%E2%80%93maximization_algorithm.
- [144] Wikipedia. F-distribution. <https://en.wikipedia.org/wiki/F-distribution>.
- [145] Wikipedia. Factor analysis. https://en.wikipedia.org/wiki/Factor_analysis.
- [146] Wikipedia. Fractal dimension. https://en.wikipedia.org/wiki/Fractal_dimension.
- [147] Wikipedia. Frisch-Waugh-Lovell theorem. https://en.wikipedia.org/wiki/Frisch%E2%80%93Waugh%E2%80%93Lovell_theorem.
- [148] Wikipedia. Functional derivative. https://en.wikipedia.org/wiki/Functional_derivative.
- [149] Wikipedia. Gamma distribution. https://en.wikipedia.org/wiki/Gamma_distribution.
- [150] Wikipedia. Gamma function. https://en.wikipedia.org/wiki/Gamma_function.
- [151] Wikipedia. Gated recurrent unit. https://en.wikipedia.org/wiki/Gated_recurrent_unit.
- [152] Wikipedia. Gene regulatory network. https://en.wikipedia.org/wiki/Gene_regulatory_network.
- [153] Wikipedia. Gibbs sampling. https://en.wikipedia.org/wiki/Gibbs_sampling.

- [154] Wikipedia. Governor device. [https://en.wikipedia.org/wiki/Governor_\(device\)](https://en.wikipedia.org/wiki/Governor_(device)).
- [155] Wikipedia. Granger causality. https://en.wikipedia.org/wiki/Granger_causality.
- [156] Wikipedia. Green's function. https://en.wikipedia.org/wiki/Green's_function.
- [157] Wikipedia. Hidden Markov model. https://en.wikipedia.org/wiki/Hidden_Markov_model.
- [158] Wikipedia. Hoeffding's inequality. https://en.wikipedia.org/wiki/Hoeffding%27s_inequality.
- [159] Wikipedia. Importance sampling. https://en.wikipedia.org/wiki/Importance_sampling.
- [160] Wikipedia. Instrumental variables estimation. https://en.wikipedia.org/wiki/Instrumental_variables_estimation.
- [161] Wikipedia. Inverse transform sampling. https://en.wikipedia.org/wiki/Inverse_transform_sampling.
- [162] Wikipedia. Jackknife resampling. [https://en.wikipedia.org/wiki/Cross-validation_\(statistics\)](https://en.wikipedia.org/wiki/Cross-validation_(statistics)).
- [163] Wikipedia. Junction tree algorithm. https://en.wikipedia.org/wiki/Junction_tree_algorithm.
- [164] Wikipedia. k-means clustering. https://en.wikipedia.org/wiki/K-means_clustering.
- [165] Wikipedia. Kalman filter. https://en.wikipedia.org/wiki/Kalman_filter.
- [166] Wikipedia. Kernel method. https://en.wikipedia.org/wiki/Kernel_method.
- [167] Wikipedia. Kernel perceptron. https://en.wikipedia.org/wiki/Kernel_perceptron.
- [168] Wikipedia. Kernel principal component analysis. https://en.wikipedia.org/wiki/Kernel_principal_component_analysis.
- [169] Wikipedia. Kramer-Kronig relations. https://en.wikipedia.org/wiki/Kramers%E2%80%93Kronig_relations.
- [170] Wikipedia. Landau pole. https://en.wikipedia.org/wiki/Landau_pole.

- [171] Wikipedia. Laplace transform. https://en.wikipedia.org/wiki/Laplace_transform.
- [172] Wikipedia. Least squares. https://en.wikipedia.org/wiki/Least_squares.
- [173] Wikipedia. Legendre transformation. https://en.wikipedia.org/wiki/Legendre_transformation.
- [174] Wikipedia. Likelihood-ratio test. https://en.wikipedia.org/wiki/Likelihood-ratio_test.
- [175] Wikipedia. Linear regression. https://en.wikipedia.org/wiki/Linear_regression.
- [176] Wikipedia. Long short term memory. https://en.wikipedia.org/wiki/Long_short-term_memory.
- [177] Wikipedia. Lotka-Volterra equations. https://en.wikipedia.org/wiki/Lotka%E2%80%93Volterra_equations.
- [178] Wikipedia. Markov blanket. https://en.wikipedia.org/wiki/Markov_blanket.
- [179] Wikipedia. Metropolis-Hastings method. https://en.wikipedia.org/wiki/Metropolis%E2%80%93Hastings_algorithm.
- [180] Wikipedia. Minimum spanning tree. https://en.wikipedia.org/wiki/Minimum_spanning_tree.
- [181] Wikipedia. Monte Carlo methods. https://en.wikipedia.org/wiki/Category:Monte_Carlo_methods.
- [182] Wikipedia. Moving-average model. https://en.wikipedia.org/wiki/Moving-average_model.
- [183] Wikipedia. Multinomial distribution. https://en.wikipedia.org/wiki/Multinomial_distribution.
- [184] Wikipedia. Multinomial theorem. https://en.wikipedia.org/wiki/Multinomial_theorem.
- [185] Wikipedia. Multivariate normal distribution. https://en.wikipedia.org/wiki/Multivariate_normal_distribution.
- [186] Wikipedia. Natural experiment. https://en.wikipedia.org/wiki/Natural_experiment.

- [187] Wikipedia. Non-deterministic finite automaton. https://en.wikipedia.org/wiki/Nondeterministic_finite_automaton.
- [188] Wikipedia. Non-negative matrix factorization. https://en.wikipedia.org/wiki/Non-negative_matrix_factorization.
- [189] Wikipedia. Ordinary least squares. https://en.wikipedia.org/wiki/Ordinary_least_squares.
- [190] Wikipedia. Petri net. https://en.wikipedia.org/wiki/Petri_net.
- [191] Wikipedia. Phase plane. https://en.wikipedia.org/wiki/Phase_plane.
- [192] Wikipedia. Pleiotropy. <https://en.wikipedia.org/wiki/Pleiotropy>.
- [193] Wikipedia. Principal component analysis. https://en.wikipedia.org/wiki/Principal_component_analysis.
- [194] Wikipedia. Program evaluation and review technique. https://en.wikipedia.org/wiki/Program_evaluation_and_review_technique.
- [195] Wikipedia. Random forest. https://en.wikipedia.org/wiki/Random_forest.
- [196] Wikipedia. Receiver operating characteristic. https://en.wikipedia.org/wiki/Receiver_operating_characteristic.
- [197] Wikipedia. Rejection sampling. https://en.wikipedia.org/wiki/Rejection_sampling.
- [198] Wikipedia. Renormalization group. https://en.wikipedia.org/wiki/Renormalization_group.
- [199] Wikipedia. Scale invariance. https://en.wikipedia.org/wiki/Scale_invariance.
- [200] Wikipedia. Score test. https://en.wikipedia.org/wiki/Score_test.
- [201] Wikipedia. Signal flow graph. https://en.wikipedia.org/wiki/Signal-flow_graph.
- [202] Wikipedia. Simple linear regression. https://en.wikipedia.org/wiki/Simple_linear_regression.
- [203] Wikipedia. Simpson's paradox. https://en.wikipedia.org/wiki/Simpson's_paradox.

- [204] Wikipedia. Spring system. https://en.wikipedia.org/wiki/Spring_system.
- [205] Wikipedia. Student's t-distribution. https://en.wikipedia.org/wiki/Student%27s_t-distribution.
- [206] Wikipedia. Support vector machine. https://en.wikipedia.org/wiki/Support-vector_machine.
- [207] Wikipedia. Survival analysis. https://en.wikipedia.org/wiki/Survival_analysis.
- [208] Wikipedia. Thermodynamics. <https://en.wikipedia.org/wiki/Thermodynamics>.
- [209] Wikipedia. Time series. https://en.wikipedia.org/wiki/Time_series.
- [210] Wikipedia. Transfer learning. https://en.wikipedia.org/wiki/Transfer_learning.
- [211] Wikipedia. Transformer (machine learning model). [https://en.wikipedia.org/wiki/Transformer_\(machine_learning_model\)](https://en.wikipedia.org/wiki/Transformer_(machine_learning_model)).
- [212] Wikipedia. Turing machine. https://en.wikipedia.org/wiki/Turing_machine.
- [213] Wikipedia. Van der pol oscillator. https://en.wikipedia.org/wiki/Van_der_Pol_oscillator.
- [214] Wikipedia. Variational Bayesian methods. https://en.wikipedia.org/wiki/Variational_Bayesian_methods.
- [215] Wikipedia. Vector autoregression. https://en.wikipedia.org/wiki/Vector_autoregression.
- [216] Wikipedia. Viterbi algorithm. https://en.wikipedia.org/wiki/Viterbi_algorithm.
- [217] Wikipedia. Wald test. https://en.wikipedia.org/wiki/Wald_test.
- [218] Wikipedia. Z-transform. <https://en.wikipedia.org/wiki/Z-transform>.
- [219] Hao Wu and Zhaohui Steve Qin. course notes, BIOS731: Advanced statistical computing, 2016 Emory Univ. <http://web1.sph.emory.edu/users/hwu30/teaching/statcomp/statcomp.html>.
- [220] Ronghui (Lily) Xu. Lecture notes, MATH 284, Spring 2020, Survival analysis. <https://mathweb.ucsd.edu/~rxu/math284/>.

- [221] Jonathan S Yedidia, William T Freeman, and Yair Weiss. Understanding belief propagation and its generalizations, Mitsubishi Technical Report tr-2001-22. <https://merl.com/publications/docs/TR2001-22.pdf>.